Statistics for Computer Scientists

Lecture 6: Estimation

17 October 2025

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Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Tba	
12 Dec	Tba	

Today's lecture

- Recap lecture 4

- Estimation
  - Point estimations
  - Intervals
- Confidence intervals for mean and proportion
- t-distribution
- Choosing sample size

Recap sampling distribution

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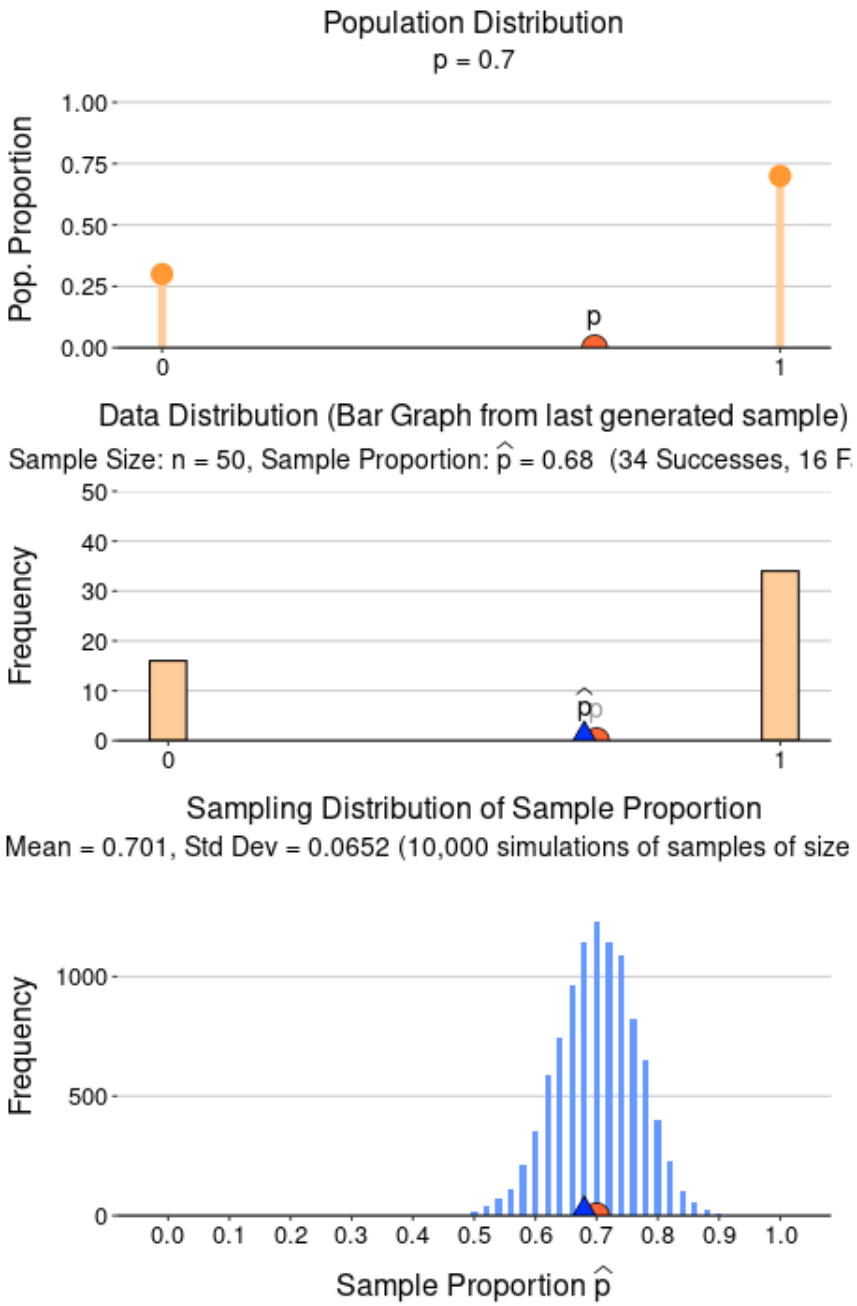
Recap

- Distribution of values of statistic, given many samples with same n
- CLT: sampling distribution approaches normal
  - Which 2 assumptions?
- Standard error: "standard deviation of the sampling distribution"
  - $se = s/\sqrt{n}$
- Z-value to probabilities (left and right tail)



Estimation: point estimates

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Statistical inference & estimation

- Goal: how do we use sample data to estimate values of population parameters?
- Point estimates: a single statistic value that is the “best guess” for the parameter value
- Interval estimates: an interval of numbers around the point estimate with a certain “confidence level” of containing the parameter value

Point estimation

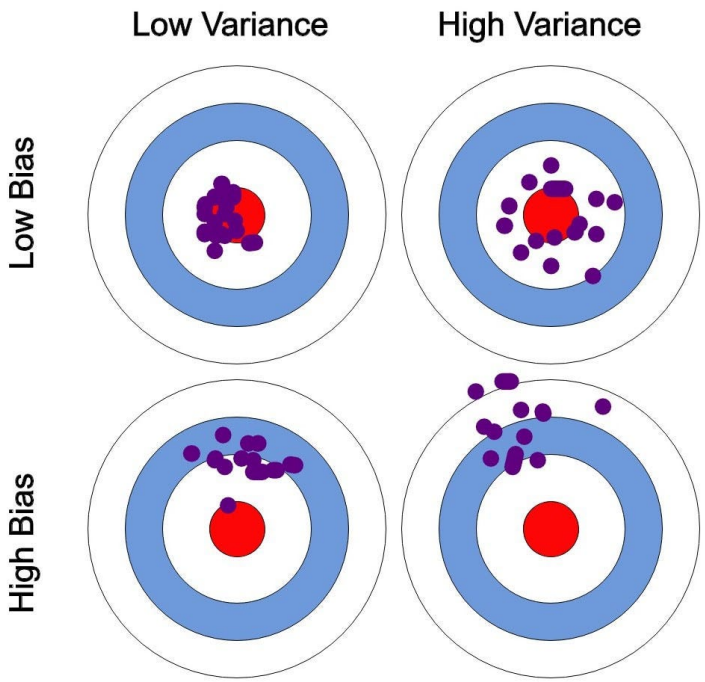
- Three common point estimators:

- **Sample mean** estimates population mean: $\hat{\mu} = \bar{y} = \frac{\sum y_i}{n}$
- **Sample standard deviation** estimates population sd: $\hat{\sigma} = s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$
- **Sample proportion** estimates population proportion: $\hat{\pi}(= p) = \frac{\#successes}{n}$

Properties of good estimators

- Unbiased: the sampling distribution of the estimator should be centered around the parameter value
  - Example of biased estimator: sample maximum – it can never be larger than the population maximum, so only negative residuals!

- Efficient: it should have the smallest possible standard error, compared to other estimators
  - Example: to estimate population mean, sample mean is more efficient than sample median



Estimation: interval estimates

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Confidence interval

- A confidence interval (CI) is an interval of numbers believed to contain the parameter value
- The **confidence level** is the probability that the interval contains the parameter (typically a number close to 1 *(or 100%)*, such as 0.95 or 0.99 *(95% or 99%)*)
- CIs are of the form **point estimate ± margin of error**
- Margin of error is based on the **standard error** and a **score**, depending on confidence level

CI for a proportion – standard error

- Consider the sample proportion $\hat{\pi}$ a mean when $Y = 1$ for the category of interest and $Y = 0$ otherwise
- The **population proportion** is the mean of a probability distribution with $P(Y = 1) = \pi$ and $P(Y = 0) = 1 - \pi$
- The **standard deviation** of this probability distribution is $\sigma = \sqrt{\pi(1 - \pi)}$
- The **standard error** of the sample proportion is:

$$\sigma_{\hat{\pi}} = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{\pi(1-\pi)}{n}}$$

CI for a proportion – score

- Remember that the **sampling distribution** for whatever population distribution will be approximately normal (CLT)
- 95% of observations fall within (1.96 * standard error) of the mean
- 99% of observations fall within (2.58 * standard error) of the mean

- So, 95% probability that $\hat{\pi}$ is between $\pi - 1.96 * \sigma_{\hat{\pi}}$ and $\pi + 1.96 * \sigma_{\hat{\pi}}$

- This would be the 95% CI for a population proportion, but…
- in practice, we don't know π, and therefore we don't know $\sigma_{\hat{\pi}}$

CI for a proportion – in practice

- The true standard error depends on an unknown parameter:
- $\sigma_{\hat{\pi}} = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{\pi(1-\pi)}{n}}$

- In practice, we estimate $\sigma_{\hat{\pi}}$ by $se = \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}}$
- And then estimate the 95% CI by $[\hat{\pi} - 1.96 * se; \quad \hat{\pi} + 1.96 * se]$

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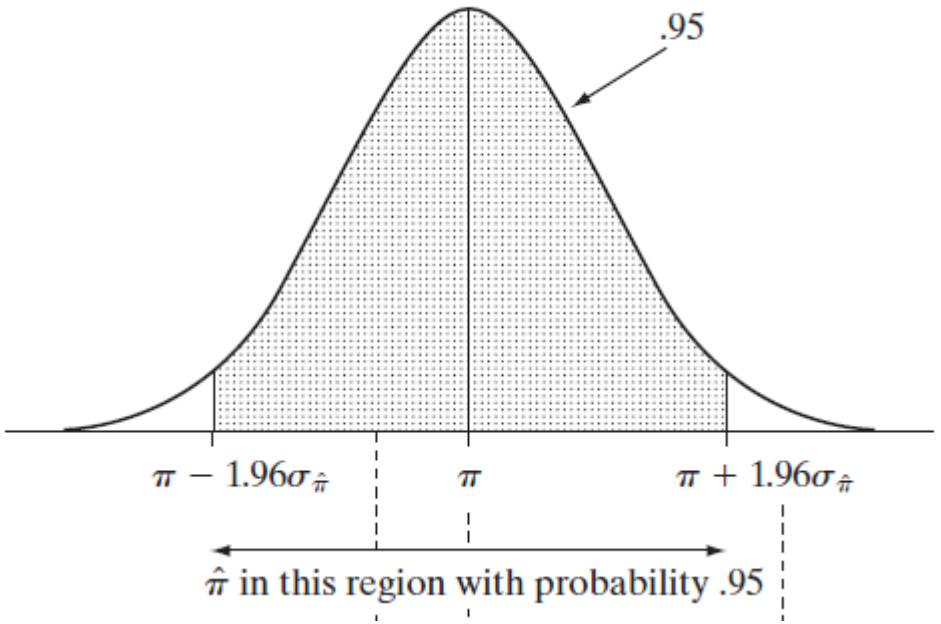
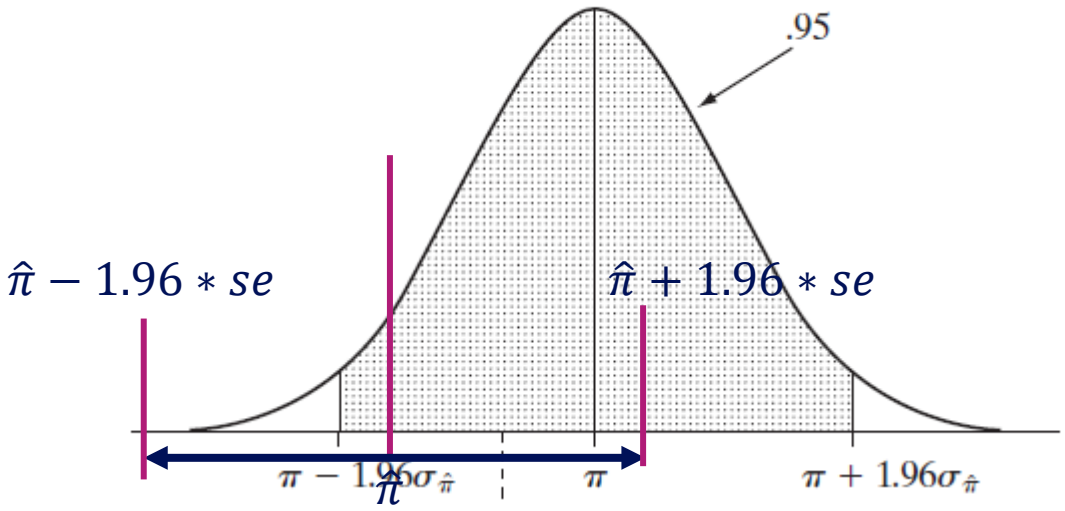
CI for a proportion – score

- Remember that the **sampling distribution** for whatever population distribution will be approximately normal (CLT)

Knowing π



Not knowing π



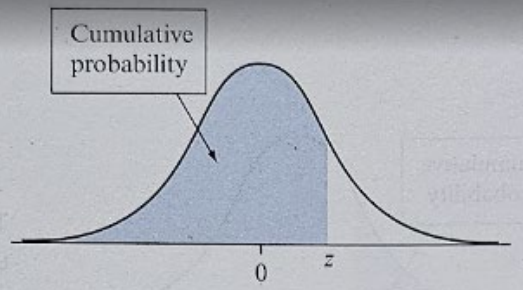
π in this region with probability .95

CI for a proportion – score

- 95% confidence:
  - 2.5% probability of being higher than interval
  - 2.5% probability of being lower than interval

- Two-sided! Look for half the "remaining" probability in the table

- Z-value for 90% confidence?



The cumulative probability for z is the area under the standard normal curve to the left of z

Table A Standard Normal Cumulative Probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
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1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817

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CI for a proportion – example

- Data says that 35 out of 164 respondents say they are "very happy"
- $\hat{\pi} = \frac{35}{164} = 0.213$ (sample proportion)
- $s = \sqrt{\hat{\pi}(1-\hat{\pi})} = 0.409$
- $se = \frac{s}{\sqrt{n}} = 0.032$

- CI is **point estimate ± margin of error**: $\hat{\pi} \pm score * se$
- 95% CI: $0.213 \pm 1.96 * 0.032 = 0.213 \pm 0.063$ *or* $(0.15, 0.28)$
- We are 95% confident that the population proportion is between 15% and 28%

CI for a proportion – example

- Data says that 35 out of 164 respondents say they are "very happy"
- 95% CI: $0.213 \pm 1.96 * 0.032 = 0.213 \pm 0.063$ *or* $(0.15, 0.28)$ (previous slide)

- 99% CI: $0.213 \pm 2.58 * 0.032 = 0.213 \pm 0.083$ *or* $(0.13, 0.30)$
  - Higher confidence level: larger score, larger margin of error, broader interval

CI for a proportion – example

- Data says that 35 out of 164 respondents say they are "very happy"
- 95% CI: $0.213 \pm 1.96 * 0.032 = 0.213 \pm 0.063$ *or* $(0.15, 0.28)$

- Data says that 350 out of 1640 respondents say they are "very happy"
- $se = \frac{s}{\sqrt{n}} = \frac{0.409}{\sqrt{1640}} = 0.01$ (was 0.032!)
- 95% CI: $0.213 \pm 1.96 * 0.01 = 0.213 \pm 0.0196$ *or* $(0.19, 0.23)$
  - Higher sample size: smaller se, smaller margin of error, narrower interval

CI for a mean

- Still of the form **point estimate ± margin of error**
- Still: margin of error = score * standard error
- Still: standard error $\sigma_{\bar{y}} = \sigma/\sqrt{n}$
- σ is now separate parameter

- Again, in practice σ is unknown, therefore we estimate $\sigma_{\bar{y}}$ by

$$se = s/\sqrt{n}$$

- (s is our best estimate of σ!)

CI for a mean – example

- Pizza delivery times are normally distributed. In a sample of 100 deliveries, we find $\bar{y} = 34$ and $s = 15$ minutes. What is the 95% confidence interval for the population mean?

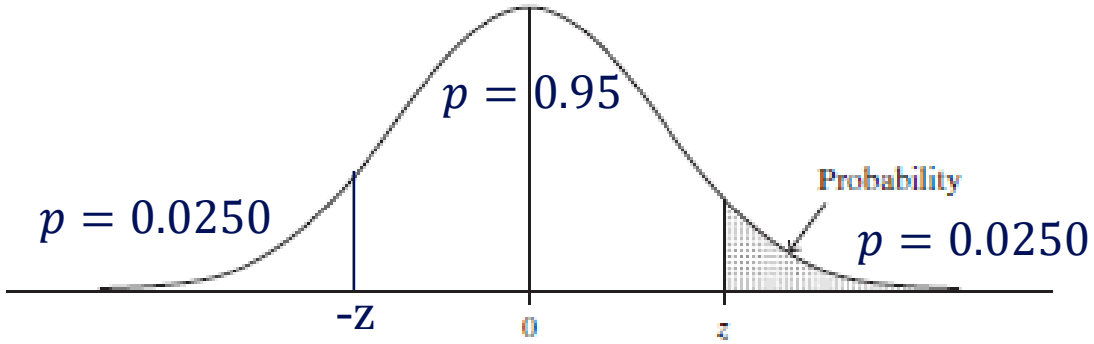
- $se = \frac{s}{\sqrt{n}} = \frac{15}{\sqrt{100}} = 1.5$
- 95% CI: $\bar{y} \pm score * se = 34 \pm 1.96 * 1.5 = 34 \pm 2.9\ or\ (31.1, 36.9)$
- We are 95% confident that the population mean of pizza delivery times is between 31.1 and 36.9 minutes

About these scores

- We used score = 1.96 for 95% CI and score = 2.58 for 99% CI
- These are z-scores
- Left-tail of –z is equal to right-tail of z
- Part in between is $1 - 2P$

	Second Decimal Place of z									
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.8	.0359	.0352	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036

TABLE A: Normal curve tail probabilities. Standard normal probability in right-hand tail (for negative values of z, probabilities are found by symmetry).



About these scores

- But if we use z-scores, then we assume the sampling distribution is normal
- When is the sampling distribution normal?

- Central limit theorem: For random sampling **with a large sample size**, the sampling distribution of the sample mean $\bar{y}$ approximates a normal distribution, no matter the shape of the population distribution
  - (large is ≥ 30)

- So, we can't use these z-scores when $n < 30$!

t-distribution

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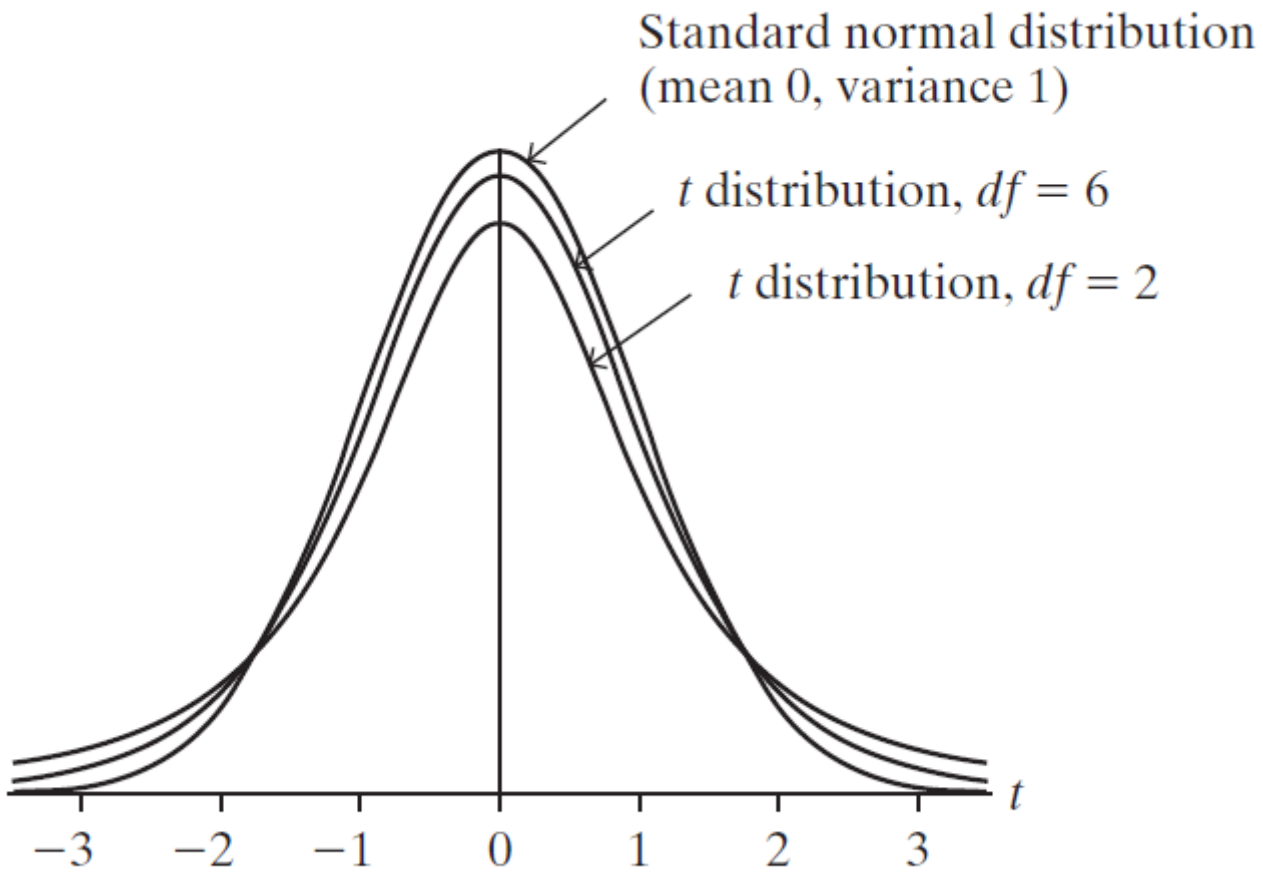
The t-distribution

- Looks like normal distribution…
  - Symmetric
  - Bell-shaped
- But it has thicker tails!
  - Meaning: extreme observations are a bit more probable
- How much thicker?
  - Depends on degrees of freedom



Degrees of freedom

- "The number of values in the final calculation that are free to vary"
- Simple example: pick a set of three numbers with a mean of 10
- Once you pick the first two numbers (say 5 and 10), the third number is not free to vary because it has to be 15 to make the mean 10
  - Two numbers are free to vary, degrees of freedom $df = 2$

- Now pick a set of ten numbers with a mean of 10: the first 9 are free to vary, the 10th is not, therefore $df = 9$
- General: $df = n - 1$

Degrees of freedom

- Degrees of freedom is $n - 1$ when we are estimating a single population mean or proportion
- Later in the course, we see other formulas for df: it all depends on how many things we are estimating

- Fully understanding df is hard and not required for this course, but know when to use which formula

t-distribution

- Smaller n means smaller df, means thicker tails
- When $df > 30$ (so $n \geq 30$), t-distribution is nearly identical to the normal distribution

Smaller n means smaller df, means thicker tails

When $df > 30$ (so $n \geq 30$), t-distribution is nearly identical to the normal distribution

- For small sample size, we must use a t-score in CI calculation rather than a z-score
- Introducing Table B!

For small sample size, we must use a t-score in CI calculation rather than a z-score

Introducing Table B!

Table A

- Remember in Table A, each z-score corresponds to a single probability; rows and columns are used for two decimal places

- But t-scores correspond to different probabilities, depending on the degrees of freedom
- We would need a 3D table to use two decimal places

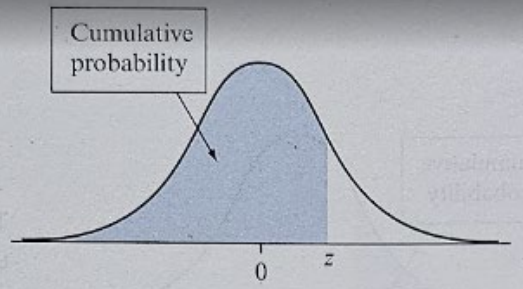


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Table B

- Rows are now used for *df*
- Columns used for common values of probability
- Cells now contain the critical (threshold) t-score, rather than the probability

	Table A	Table B
Rows	z-score (1st decimal)	df
Columns	z-score (2nd decimal)	probability
Cells	probability	t-score

	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
	Right-Tail Probability					
df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
1	3.078	6.314	12.706	31.821	63.656	318.289
2	1.886	2.920	4.303	6.965	9.925	22.328
3	1.638	2.353	3.182	4.541	5.841	10.214
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.894
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.611
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.261

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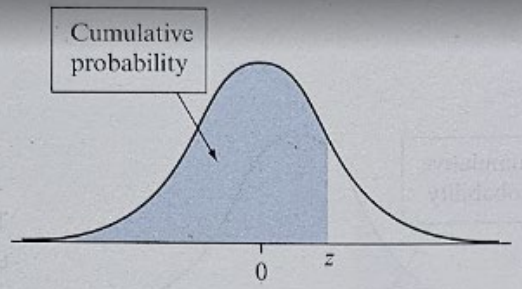

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0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817

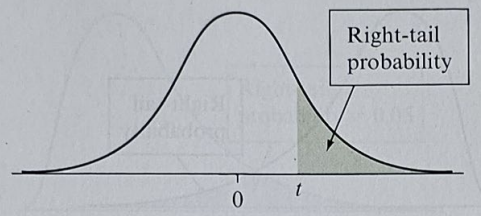

Table B t Distribution Critical Values

	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
	Right-Tail Probability					
df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
1	3.078	6.314	12.706	31.821	63.656	318.289
2	1.886	2.920	4.303	6.965	9.925	22.328
3	1.638	2.353	3.182	4.541	5.841	10.214
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.894
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
100	1.290	1.660	1.984	2.364	2.626	3.174
∞	1.282	1.645	1.960	2.326	2.576	3.091

CI for a population mean

- Pizza delivery times are normally distributed. In a sample of 10 deliveries, we find $\bar{y} = 34$ and $s = 15$ minutes. What is the 95% confidence interval for the population mean?
- Only 10 deliveries! We cannot assume a normal sampling distribution
- $se = \frac{s}{\sqrt{n}} = \frac{15}{\sqrt{10}} = 4.7$
- 95% CI: $\bar{y} \pm score * se$

- Which t-score?

Table B

- 95% CI… right-tail probability is 0.025
- $df = n - 1 = 9$
- $t = 2.262$

- Quite a bit larger than z-score would be! (1.96)

	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
	Right-Tail Probability					
df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
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14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.611
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.261
60	1.296	1.671	2.000	2.390	2.660	3.232
80	1.292	1.664	1.990	2.374	2.639	3.195
100	1.290	1.660	1.984	2.364	2.626	3.174
∞	1.282	1.645	1.960	2.326	2.576	3.091

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CI for a population mean

- Pizza delivery times are normally distributed. In a sample of 10 deliveries, we find $\bar{y} = 34$ and $s = 15$ minutes. What is the 95% confidence interval for the population mean?
- Only 10 deliveries! We cannot assume a normal sampling distribution
- $se = \frac{s}{\sqrt{n}} = \frac{15}{\sqrt{10}} = 4.7$
- 95% CI: $\bar{y} \pm score * se = 34 \pm 2.262 * 4.7 = 34 \pm 10.6 = (23.4, 44.6)$

α

- Confidence interval: if we repeatedly take random samples of size n and each time calculate a 95% CI, in the long run, we expect that 95% of those CI's contain the true population proportion/mean
- The probability that the CI **does not** contain the true parameter is called the error probability, denoted by α
- In the columns of Table B, the subscript shows $\frac{1}{2}\alpha$

Table B *t* Distribution Critical Values

	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
	Right-Tail Probability					
df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
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2	1.886	2.920	4.303	6.965	9.925	22.328
3	1.638	2.353	3.182	4.541	5.841	10.214

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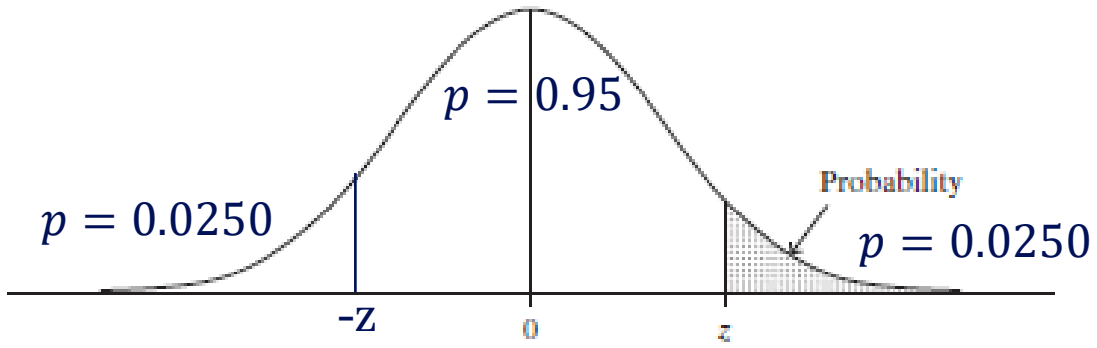
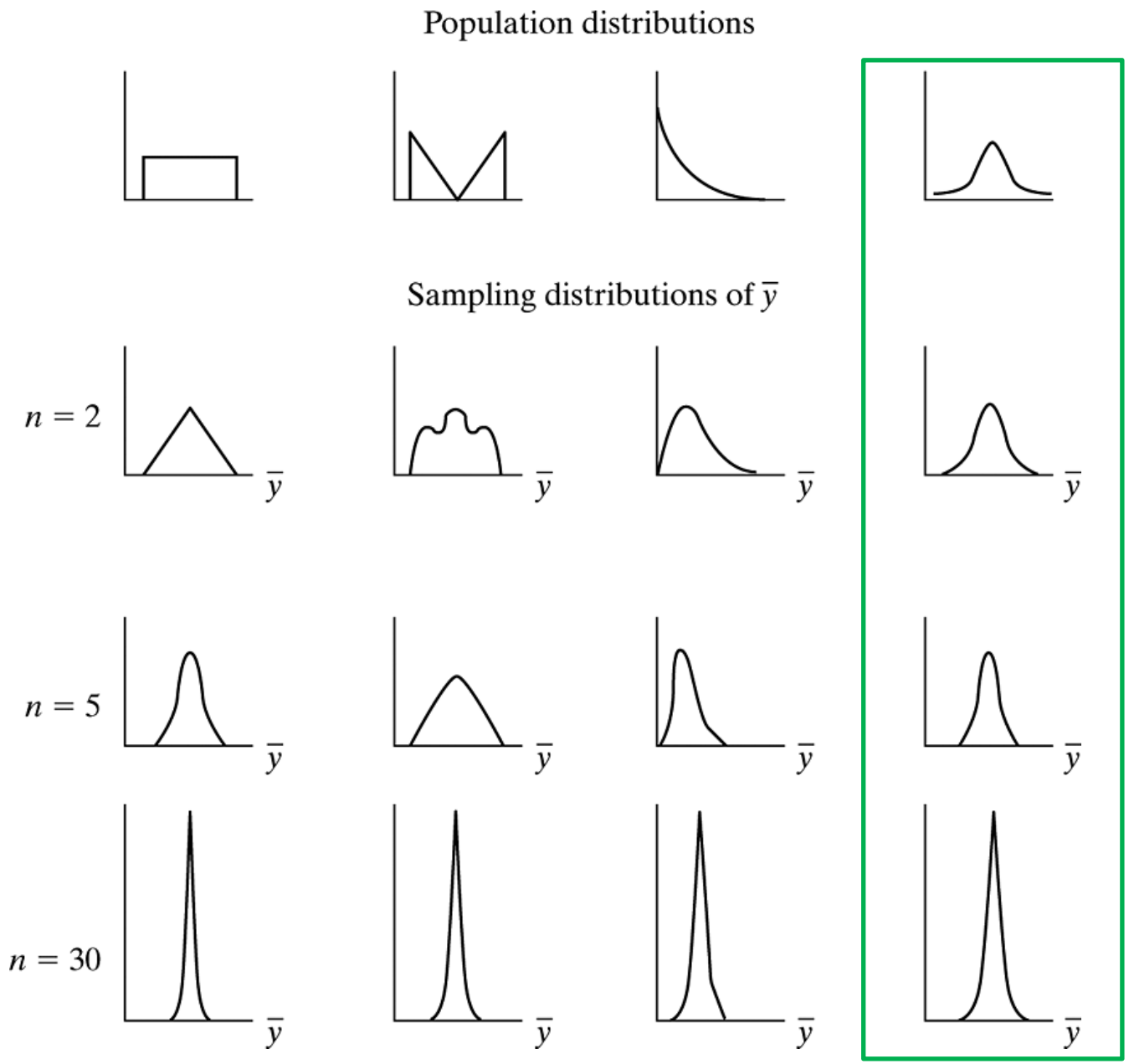
TABLE A: Normal curve tail probabilities. Standard normal probability in right-hand tail (for negative values of z, probabilities are found by symmetry).



Which score to use for estimating CI

- When estimating a mean:
  - Use a t-score
  - But, if $n \geq 30$, you may use the z-score instead, as it is almost identical

- When estimating a proportion:
  - Use a z-score
  - If too small sample size, we don't estimate a CI! t-distribution only applies to means, not proportions (why?)


Population distributions
Sampling distributions of $\bar{y}$
$n = 2$
$n = 5$
$n = 30$
$\bar{y}$

Some comments on CI for means

- This method, using t-scores, is robust to violations of the assumption of a normal population distribution
- However, if the sample distribution is highly skewed, it may not give accurate CIs

- For both means and proportions:
  - Greater confidence requires wider CI
  - Greater sample size produces narrower CI

Choosing sample size

Universiteit Leiden
The Netherlands

Choosing sample size – proportion

- We've used this formula to calculate the margin of error (assume a 95% CI):

$$\text{Margin of error} = 1.96 * se = 1.96 * \sqrt{\hat{\pi}(1-\hat{\pi})/n}$$

- How large a sample size do we need to estimate the population proportion with a margin of error of 0.03, with 95% probability?

$$1.96 * \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}} = 0.03, \text{ rewrite to } n = \hat{\pi}(1-\hat{\pi})\left(\frac{1.96}{0.03}\right)^2 = 4268\hat{\pi}(1-\hat{\pi})$$

- So it depends on $\hat{\pi}$, but we can see that $\hat{\pi}(1-\hat{\pi})$ is largest for $\hat{\pi} = 0.5$
- To be safe, use a sample size of $n = 4268 * 0.5 * 0.5 = 1067$

Choosing sample size – proportion

- To be safe, use $\hat{\pi} = 0.5$ to estimate sample size (will give largest n)
- But if we have an educated guess, use that!

- Example: a previous study suggests that the population proportion is roughly 0.20. To get a margin of error or 0.03 for a 95% CI:

$$n = \hat{\pi}(1 - \hat{\pi})\left(\frac{1.96}{0.03}\right)^2 = 0.2 * 0.8 * \left(\frac{1.96}{0.03}\right)^2 = 683$$

- It is "easier" to estimate population proportions close to 0 or 1, than those around 0.5

Choosing sample size – mean

- For a mean, we know that $se = \frac{s}{\sqrt{n}}$ and margin of error = $1.96 * s/\sqrt{n}$
- How large a sample size do we need to estimate the population mean with a margin of error of 5, with 95% probability?
- We need a guess for the value of s!

- $n = s^2 * \left(\frac{1.96}{5}\right)^2$

Choosing sample size – mean – example

- Sushi delivery times are approximately normally distributed. In a previous study, we found that $\bar{y} = 50$, $s = 8$ minutes with $n = 35$.

- This gives a 95% CI of (47.3, 52.7) minutes (margin of error = 2.7 minutes)
- Which n would we need to get a margin of error of 1 minute in our next study?

- Now we have an estimate for s!
- $n = 8^2 * \left(\frac{1.96}{1}\right)^2 = 246$

General formulas

- For proportions:

$$n = \pi(1-\pi)\left(\frac{z}{M}\right)^2 \quad \text{(in practice, } \pi \text{ is estimated by } \hat{\pi}\text{)}$$

- For means:

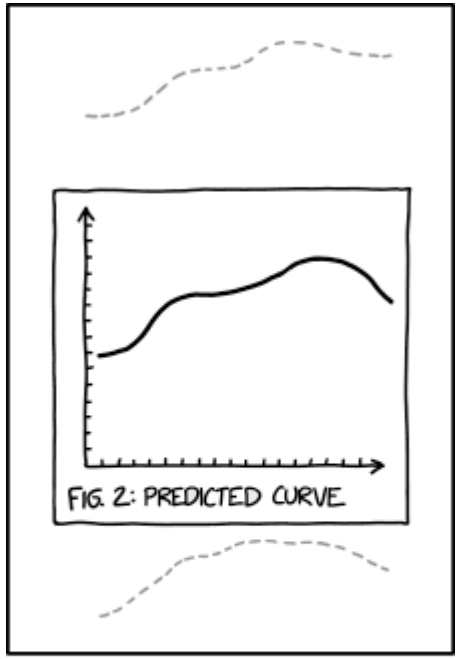
$$n = \sigma^2\left(\frac{z}{M}\right)^2 \quad \text{(in practice, } \sigma^2 \text{ is estimated by } s^2\text{)}$$

- With z being the z-score corresponding to the chosen confidence level (e.g., 1.96 for 95%), and M being the desired margin of error

Wrapping up

- Read chapter 8 of the book
- Tutorial: confidence intervals in JASP

- Next week: no class
- 31 Oct: Significance testing!
  - Lectures by Matthijs



SCIENCE TIP: IF YOUR MODEL IS BAD ENOUGH, THE CONFIDENCE INTERVALS WILL FALL OUTSIDE THE PRINTABLE AREA.

https://xkcd.com/2311/